

SPEAKER-GROUP SENSITIVITY OF PAIRED DETECTOR COMPARISONS ON ASVspoof 2021 DF

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ABSTRACT

We examine how the composition of an evaluation set affects small differences in equal-error rate (EER) between speech deepfake detectors. Using public scores from eight ASVspoof 2021 DF systems, we compare resampling individual trials with resampling observed speakers and spoof attacks, keeping detector comparisons paired and refitting EER thresholds. Across all 28 system pairs, jointly constructed sensitivity bands exclude zero for 26 pairs under trial resampling and 18 under speaker-attack resampling. Speaker-only resampling gives the same decisions. Removing one group of 315 bona-fide trials reduces the XLSR-Mamba advantage over XLSR-Conformer from 0.389 to 0.129 percentage points. All sixteen comparisons between the four self-supervised systems and four organizer baselines remain separated. A limited SpoofCeleb analysis gives a different speaker-versus-attack pattern. These results motivate reporting sensitivity to observed evaluation groups; they do not establish population coverage or a universal resampling prescription.

Index Terms— Audio deepfake detection, paired evaluation, speaker influence, EER

1. INTRODUCTION

On ASVspoof 2021 DF, XLSR-Mamba achieves 1.884% EER and XLSR-Conformer 2.273%. Deleting VCC2SM3, which contains 315 bona-fide VCC2018 trials and no spoof trials, changes the Mamba-Conformer gap from -0.389 to -0.129 percentage points. Those 315 trials are part of 14,869 bona-fide trials in a benchmark with 533,928 trials overall. The gap remains negative, but its magnitude depends strongly on one observed group.

This deletion belongs to a complete family analysis, with all speakers retained in the principal comparison. Across eight detectors, we ask which paired EER differences survive changing the resampling unit, and how concentrated the delete-one speaker influence is in the affected modern comparisons. ASVspoof 2021 DF has 93 speakers and 110 attacks; a large trial count need not provide broad group support for a close comparison. Shared group variation may cancel between detectors, so dependence of individual scores alone cannot answer this paired question.

Our contribution is an empirical concentration result on fixed scores: speaker-only resampling reproduces the joint decisions, and delete-one influence is concentrated to differing degrees across the three affected self-supervised (SSL) comparisons. We print every primary trial and joint band and repeat the analysis within the six SSL comparisons. A limited SpoofCeleb counterexample retains the reason to examine both factors. The reporting consequence is to disclose influential-group sensitivity when claiming a small paired EER advantage.

2. RELATED WORK

Benchmark reporting. ASVspoof 2017 reported EER intervals [1]; ASVspoof 2021 Fig. 4(c) used the Bengio-Mariéthoz test [2] with Holm correction [3]. Its shared-trial variant accounts for two systems using the same trials, not repeated speakers and attacks. We found no uncertainty analysis in the later journal overview [4] or ASVspoof 5 [5]; a recent SSL benchmark reports point EERs [6]. Other work reports trial-bootstrap ranks [7], EER spreads [8], corrected pairwise tests [9], or fixed-threshold intervals [10]. Training-seed variation also moves rankings [11]; our scores hold training fixed.

Dependent paired comparison. User-aware biometric and speech resampling [12, 13], correlation-corrected paired tests [14], and synchronized resampling for correlated classifiers [15] precede this study. Our speaker and spoof-attack factors are crossed [16, 17, 18], distinct from the nested-observation designs studied in [19]. We use these established tools to measure concentrated group influence on paired EER comparisons, with every threshold refitted. Recent audio-deepfake work also reports paired and speaker-resampled EER-difference intervals [20]; here we examine a complete comparison family and the concentration of speaker-deletion influence.

Evaluation design. Aggregation weights [21], channel presentation [22], representations [23], controlled generator/speaker perturbations [24] and corpus overlap [25] affect deepfake evaluation. Identity perturbations [26], speaker-

adversarial training [27] and speaker/synthesizer familiarity [28] address related sensitivities. Here the scores are fixed and evaluation weights vary. Simultaneous rank intervals are established [29]. Because selecting only neighbors in the observed point-EER ordering uses the outcomes, we include every detector pair.

3. METHOD

3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, the false-reject rate (FRR) is the fraction of bona-fide weight at or below it and the false-accept rate (FAR) the fraction of spoof weight above it; we select the first position minimizing their absolute difference and take their mean as EER, without interpolation. For ordered pair (A, B) , $\Delta\text{EER}_{A,B} = 100(\text{EER}_A - \text{EER}_B)$ percentage points (negative favors A); every replicate refits both thresholds. A pair is *separated under procedure h* when its simultaneous band excludes zero; write this indicator as $I_{p,h}$.

3.2. Perturbation laws and scope

Groups use the speaker-ID and attack-ID fields in the official DF metadata. For converted speech, the speaker ID denotes the target voice; it need not identify the source recording’s speaker.

The trial bootstrap independently resamples trials within class. For SpoofCeleb only, it samples 91,130 trial indices with replacement from the pooled list, so class counts vary across replicates. In the speaker $\times$ attack product-weight (PW) bootstrap [18, 30], we independently draw speakers and spoof attacks uniformly with replacement, drawing as many of each as observed; a spoof trial receives the product of its speaker and attack multiplicities, and a bona-fide trial receives only its speaker multiplicity. Speaker-only fixes attack multiplicities at one; attack-only fixes speaker and bona-fide trial weights at one. Each of these four procedures uses 5,000 replicates. Each replicate’s weights are shared across systems; procedures have separate random streams.

Interpretation. These bands describe sensitivity of fixed scores to the stated weights, not confidence about future speakers or attacks. Four VCC spoof strata contain only 4, 4, 4 and 6 speakers. Coverage results for crossed-array means do not establish coverage for refitted EER differences [30]. The one-factor, Monte Carlo, mean/sign, pairing, tie and group-deletion diagnostics were designed after the primary results and are exploratory. Matching separation decisions does not decompose variance or identify a causal speaker effect. A band containing zero establishes neither equality nor a deployment ranking.

3.3. All-pair simultaneous bands

For procedure h , let $s_{p,h} = \text{sd}_b(\Delta_{p,h}^{(b)})$. We set $q_{.95,h}$ to the empirical 95th percentile of $\max_{p=1}^{28} |\Delta_{p,h}^{(b)} - \bar{\Delta}_{p,h}|/s_{p,h}$ and report $\hat{\Delta}_p \pm q_{.95,h} s_{p,h}$ [31]. The quantile centers deviations on the bootstrap mean, whereas the bands center on the original estimate; band inclusion is not an empirical reversal probability. The maximum includes all 28 primary contrasts, including when reporting within-cohort counts; SpoofCeleb uses its own complete six-pair family. No population error-control guarantee is asserted.

3.4. Delete-one concentration

For the delete-one analysis, equal scores are ordered by trial ID. For each of the 93 speakers, we remove all its trials, renormalize within class and refit both thresholds. For pair p , define $d_{p,g} = \hat{\Delta}_{p,-g} - \bar{\Delta}_{p,-\cdot}$, where the bar averages the 93 delete-one estimates. The influence share is $d_{p,g}^2 / \sum_{h=1}^{93} d_{p,h}^2$; top-five shares sum the five largest terms of this same denominator. These are shares of squared deletion deviations, not fractions of the original gap or population effective sample sizes.

4. RESULTS

Setup. The primary set contains XLSR-Mamba [32], XLSR+SLS (SLS below) [33], XLSR-Conformer [34] and SSL-AASIST [35], plus RawNet2, LFCC-LCNN, LFCC-GMM and CQCC-GMM. All 533,928 evaluation-phase trials have every score: 14,869 bona-fide and 519,059 spoof trials, with weights normalized within class. The matched trial/PW comparison uses 5,000 replicates and seed 2026081604; NumPy SeedSequence children are trial, joint, speaker-only and attack-only in that order.

Complete paired evidence. The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Table 1 prints all point differences and both simultaneous bands. Trial and joint bands separate 26/28 and 18/28 pairs; within SSL, the counts fall from 5/6 to 2/6. All primary within-SSL losses involve Conformer. All sixteen SSL-versus-baseline comparisons remain separated under every primary arm (Table 3). Restricting the maximum to the six SSL pairs also gives 5/6 trial and 2/6 joint separations; the losses do not require the organizer comparisons in the family.

Concentration across affected SSL pairs. Table 2 reports the largest speaker contributor for each SSL comparison losing separation. The two comparisons of Mamba and SLS with Conformer share VCC2SM3 as their largest contributor, whereas Conformer-SSL-AASIST has a different and less dominant contributor. All speakers remain in the principal analysis. Deletion changes class normalization and refits

Table 1. Complete primary 21DF sensitivity bands in percentage points, using the full 28-pair maximum. Pair labels mean first minus second; negative favors the first. A star marks trial separation lost under joint resampling. Abbreviations and point EERs (%) are defined above the bands.

M	XLSR-Mamba	1.884	R	RawNet2	22.383
S	XLS-R+SLS	1.916	L	LFCC-LCNN	23.477
C	XLSR-Conformer	2.273	F	LFCC-GMM	25.247
A	SSL-AASIST	2.852	Q	CQCC-GMM	25.563

Pair	Gap	Trial band	Joint band
Within SSL			
M-S	-0.032	[-0.144, 0.080]	[-0.725, 0.661]
M-C*	-0.389	[-0.570, -0.208]	[-1.348, 0.570]
M-A	-0.968	[-1.175, -0.760]	[-1.828, -0.107]
S-C*	-0.357	[-0.538, -0.177]	[-1.493, 0.779]
S-A	-0.935	[-1.141, -0.730]	[-1.784, -0.087]
C-A*	-0.578	[-0.835, -0.322]	[-1.637, 0.480]
Within organizer baselines			
R-L*	-1.094	[-1.722, -0.466]	[-5.528, 3.341]
R-F*	-2.864	[-3.654, -2.074]	[-9.949, 4.221]
R-Q*	-3.180	[-4.003, -2.356]	[-9.284, 2.925]
L-F*	-1.770	[-2.426, -1.114]	[-6.490, 2.950]
L-Q*	-2.086	[-2.765, -1.407]	[-6.173, 2.001]
F-Q	-0.316	[-0.929, 0.297]	[-3.370, 2.738]
SSL versus organizer baselines			
M-R	-20.499	[-21.070, -19.929]	[-26.950, -14.049]
M-L	-21.593	[-22.123, -21.063]	[-27.445, -15.742]
M-F	-23.363	[-23.981, -22.746]	[-29.462, -17.264]
M-Q	-23.679	[-24.364, -22.994]	[-29.247, -18.112]
S-R	-20.467	[-21.039, -19.896]	[-26.958, -13.976]
S-L	-21.561	[-22.090, -21.032]	[-27.440, -15.682]
S-F	-23.331	[-23.948, -22.714]	[-29.469, -17.194]
S-Q	-23.647	[-24.330, -22.964]	[-29.216, -18.078]
C-R	-20.110	[-20.673, -19.547]	[-26.307, -13.914]
C-L	-21.204	[-21.726, -20.682]	[-26.757, -15.651]
C-F	-22.974	[-23.587, -22.361]	[-28.899, -17.049]
C-Q	-23.290	[-23.964, -22.616]	[-28.709, -17.871]
A-R	-19.532	[-20.118, -18.945]	[-25.739, -13.325]
A-L	-20.626	[-21.171, -20.080]	[-26.215, -15.036]
A-F	-22.396	[-23.028, -21.763]	[-28.399, -16.392]
A-Q	-22.712	[-23.406, -22.017]	[-28.123, -17.301]

Table 2. Speaker deletion influence for the three affected SSL pairs (abbreviations as in Table 1). Gaps are in points; shares use the squared-deviation denominator in Method.

Pair	Largest group	Full → deletion	Largest / top five
M-C	VCC2SM3	-0.389 → -0.129	77.09% / 90.03%
S-C	VCC2SM3	-0.357 → -0.131	55.85% / 78.47%
C-A	VCC2SF2	-0.578 → -0.441	23.85% / 75.74%

thresholds; it measures influence without isolating a causal speaker effect.

Factor pattern and pairing. Speaker-only reproduces

Table 3. Primary zero-exclusion counts under the full 28-pair correction; every arm retains 16/16 SSL-versus-baseline separations. Attack ranges span the original and five fresh 5,000-draw runs; other entries are recorded-run counts. One-factor analyses are exploratory.

Roster / subset	Trial	Speaker	Attack	Joint
21DF, all	26/28	18/28	21-22/28	18/28
21DF, SSL	5/6	2/6	3/6	2/6
21DF, organizer	5/6	0/6	2-3/6	0/6

Table 4. Mamba minus Conformer: mean shift from the point gap, pair SD (points), opposite-sign draws out of 5,000, and zero exclusion under the full 28-pair band.

Law	Mean shift	SD	Opposite	Excludes?
Trial	-0.002492	0.061488	0	Yes
Speaker	-0.003175	0.270826	210	No
Attack	-0.010235	0.147333	37	No
Joint	-0.010135	0.319971	463	No

every primary joint separation decision; attack-only reproduces four or five of the eight joint losses across Monte Carlo repeats. Matching indicators does not make attacks inert: Table 4 shows different SDs and opposite-sign counts under speaker-only and joint resampling. Under the joint law, Mamba and Conformer EERs correlate 0.777; pairing removes about 75% of the sum of their marginal variances, yet the paired SD rises from 0.06149 under trials to 0.31997 points under joint resampling. The trial and joint critical values are 2.940 and 2.997, so SD inflation drives the band widening. The joint mean displacement is only 0.032 SD, so band centering does not explain inclusion of zero; the trial band excludes zero with 0/5,000 opposite-sign draws, the joint band includes it with 463/5,000. A finite zero count does not establish zero reversal probability.

Numerical checks. Across five fresh attack-only runs, separation counts range from 21 to 22 overall and from two to three among organizer baselines; within-SSL counts remain three. Five fresh 5,000-draw runs for each of the trial, speaker-only and joint arms also preserve every recorded separation indicator. A distinct-score-boundary tie check over the four primary arms and five fresh attack streams changes no indicator; the largest pair-band endpoint change is 0.000233 points.

Counterexample on single-source SpoofCeleb. On all 91,130 SpoofCeleb evaluation trials [36], pooled trial resampling separates 6/6 pairs and joint resampling 3/6; class-stratified trial resampling also gives 6/6. EERs are 57.93%, 24.51%, 26.72% and 27.58% for AASIST, SLS, SSL-AASIST and Mamba. The roster has 40 speakers and nine attacks. Trial/joint arms use seed 20260829; the post-result one-factor check uses seed 20260830. Every base item

Table 5. SpoofCeleb: all six pairs under their own six-pair maximum. T/S/A/J denote trial/speaker/attack/joint; Y means the band excludes zero, N means it includes zero. SSL-A denotes SSL-AASIST. Below: point gaps and trial/joint bands (points) for the three joint losses.

Pair	T	S	A	J
AASIST-SLS	Y	Y	Y	Y
AASIST-SSL-A	Y	Y	Y	Y
AASIST-Mamba	Y	Y	Y	Y
SLS-Mamba	Y	Y	N	N
SSL-A-SLS	Y	Y	N	N
SSL-A-Mamba	Y	N	N	N
Separated	6/6	5/6	3/6	3/6

Pair	Gap	Trial	Joint
SLS-Mamba	-3.063	[-3.771, -2.355]	[-16.062, 9.936]
SSL-A-SLS	2.207	[1.592, 2.821]	[-7.392, 11.806]
SSL-A-Mamba	-0.856	[-1.446, -0.266]	[-14.805, 13.093]

has one bona-fide and nine spoof renditions under one source label, so between-source mass cannot change. Under the original six-pair correction, all three comparisons among SLS, SSL-AASIST and Mamba lose separation; the three AASIST contrasts retain it. Table 5 prints the four-arm decisions and bands for the joint losses. Here attack-only reproduces the joint indicator vector, while speaker-only separates 5/6 pairs. This is a control on a weakly transferred, off-domain roster, not evidence that attack effects generally dominate speaker effects on another corpus. Different rosters and poor transfer prevent a causal attribution of this contrast to the corpus; the mirrored factor pattern nevertheless counters a universal speaker-only prescription.

5. DISCUSSION AND CONCLUSION

The contribution is measured concentration across paired comparisons on this fixed benchmark, not a new confidence procedure. A diffuse delete-one profile across the affected SSL pairs would contradict the concentration claim; Table 2 exposes both the strong flagship case and the weaker third pair. A changed speaker-only decision would contradict the reported primary factor agreement. Neither finding establishes why those groups matter, and SpoofCeleb shows why the factor pattern should not become a universal prescription.

For a small paired EER advantage on this benchmark, report the point difference, trial and group bands for the same family, and influential-group deletion sensitivity. If only the trial band excludes zero, describe the advantage as sensitive to evaluation composition and seek broader group support before claiming robust superiority. Publish the group memberships and weighting rules needed to reproduce the check. The sixteen SSL-versus-baseline contrasts survive all four

primary resampling laws, while close within-cohort comparisons are sensitive. These results support reporting sensitivity to observed-group weights, without establishing population confidence or a universal factor pattern across corpora.

Artifact. The public artifact provides the primary-analysis code, score provenance, numerical results and the additional checks identified in its reproduction guide. Artifact version `ef1be20cd8f6de8006c10a4dc04983ec96626af1` at github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzed previously collected ASVspoof 2021 and SpoofCeleb recordings and detector scores. It involved no new recording or interaction with human participants; no ethical approval was required. The authors declare no competing interests.

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